

ABOUT THIS BOOK

Animal is organized into 3 main sections: a general **Introduction** to animals and their lives; a section on the world's main **Habitats**; and the main part of the book, **The Animal Kingdom**, which is dedicated to the description of animal groups and species. At the end of the book, a detailed glossary defines all zoological and technical terms used, while a full index lists all the groups and species featured, by both Latin name and common name, including alternative common names.

INTRODUCTION

Animal begins with an overview of all aspects of animal life. This includes an account of what an animal is—and how it differs from other living things. It also examines animal anatomy, life cycles,

evolution, behavior, and conservation. In addition, there is a comprehensive presentation of the classification scheme that underpins the species profiles in **The Animal Kingdom**.

ANATOMY

All animals are made up of various parts, each with a specific function. The smallest functional parts are cells, which are organized into tissues, organs, and systems. The structure of these systems varies between one type of animal and another, and also between animals that live in different ways, but the way they work is the same.

Body systems

The body systems of an animal are the organs and tissues that work together to perform the functions of the body. These include the digestive system, the circulatory system, the respiratory system, the excretory system, the reproductive system, and the nervous system.

Support and movement

Animals have different ways of supporting their bodies and moving. Some have skeletons, some have shells, and some have no external support at all. The way an animal moves depends on its body structure and its environment.

Symmetry

Many animals have a body plan that is symmetrical, meaning that one side of the body is a mirror image of the other. This is called bilateral symmetry. Some animals, like jellyfish, have radial symmetry, and some, like sponges, have no symmetry at all.

Reproduction

Animals reproduce in different ways. Some lay eggs, some give birth to live young, and some have a mix of both. The way an animal reproduces depends on its life cycle and its environment.

Evolution

All animals are related, and they have evolved from a common ancestor. The changes that have occurred over time are called evolution. These changes are caused by natural selection, which is the process by which organisms with traits that are better suited to their environment survive and reproduce.

Classification

Animals are classified into groups based on their characteristics. The classification system is called taxonomy, and it is used to organize the vast number of animal species into a manageable format.

Conservation

Many animal species are at risk of extinction, and it is important to take steps to protect them. Conservation efforts include protecting habitats, reducing pollution, and breeding endangered species in captivity.

HABITATS

This section looks at the world's main animal habitats. Coverage of each habitat is divided into 2 parts. The first (illustrated below) describes the habitat itself, including its climate and plant

life, and the types of animals found there. The pages that immediately follow describe how the anatomy and behavior of animals are adapted to suit the conditions in which they live.

general introduction to habitat type

description of more specific habitat sub-type

dotted lines identify distinct zones within habitat

Mountains

Mountains are large landforms that rise above the surrounding land. They are formed by tectonic forces and can be found in all parts of the world. Mountains have a unique climate and are home to many rare and endangered species.

Temperate mountains

Temperate mountains are found in the temperate zones of the world. They have a climate with four distinct seasons and are home to a variety of plants and animals.

Tropical mountains

Tropical mountains are found in the tropical zones of the world. They have a hot, wet climate and are home to a variety of plants and animals.

High-altitude mountains

High-altitude mountains are found in the high-altitude zones of the world. They have a cold, dry climate and are home to a variety of plants and animals.

map showing global distribution of habitat

photographs show representative animals found within each zone

feature on conservation issues

THE ANIMAL KINGDOM

This section of *Animal* is divided into 6 chapters: mammals, birds, reptiles, amphibians, fishes, and invertebrates. Each chapter begins with an introduction to the animal group. Lower-ranked groups such as orders and families are then introduced, followed by profiles of the species classified within them. The invertebrates are organized slightly differently (see opposite), while passerine birds are profiled at family level, with only representative species shown.

HABITAT SYMBOLS

Symbols in profiles are listed as shown below, not in order of main occurrence.

- Temperate forest, including woodland
- Coniferous forest, including woodland
- Tropical forest and rain forest
- Mountains, highlands, scree slopes
- Desert and semidesert
- Open habitats, including grassland, moor, heath, savanna, fields, scrub
- Wetlands and all still bodies of water, including lakes, ponds, pools, marshes, bogs, and swamps
- Rivers, streams, and all flowing water
- Mangrove swamps, above or below the waterline
- Coastal areas, including beaches and cliffs, areas just above high tide, in the intertidal zone, and in shallow, offshore waters
- Seas and oceans
- Coral reefs and waters immediately around them
- Polar regions, including tundra and icebergs
- Urban areas, including buildings, parks, and gardens
- Parasitic; living on or inside another animal

DATA FIELD

Summary information is given at the start of each profile. Measurements are for adult males of the species and may be a typical range, single-figure average, or maximum, depending on available records.

- LENGTH** (all groups, except Invertebrates)
MAMMALS: head and body. **BIRDS:** tip of bill to tip of tail. **REPTILES:** length of carapace for tortoises and turtles; head and body, including tail, for all other species. **FISHES & AMPHIBIANS:** head and body, including tail. **TAIL** (Mammals only) Length.
- WEIGHT** (Mammals, Birds, and Fishes only)
Body weight.
- SOCIAL UNIT** (Mammals only)
Whether a species lives mainly alone (*Individual*), in a group, in a *Pair*, or varies between these units (*Variable*).
- PLUMAGE** (Birds only)
Whether sexes are *alike* or *different*.
- MIGRATION** (Birds only)
Whether a bird is a *Migrant*, *Partial migrant*, *Nonmigrant*, or *Nomadic*.
- BREEDING** (Reptiles and Fishes only)
Whether the species is *Viviparous*, *Oviparous*, or *Ovoviviparous*.
- HABIT** (Reptiles and Amphibians only)
Whether the species is partly or wholly *Terrestrial*, *Aquatic*, *Burrowing*, or *Arboreal*.
- BREEDING SEASON** (Amphibians only)
The time of year in which breeding occurs.

- SEX** (Fishes only)
Whether the species has separate males and females (*Male/Female*), is a *Hermaphrodite*, or a *Sequential hermaphrodite*.
- OCCURRENCE** (Invertebrates only)
Number of species in the family, class, or phylum; their distribution and the microhabitats they can be found in.
- STATUS** (all groups)
Animal uses The IUCN Red List (see p.33) and other threat categories, as follows:
EXTINCT IN THE WILD (IUCN) Known only to survive in captivity, as a naturalized population outside its natural range, or as a small, managed population that has been reintroduced.
CRITICALLY ENDANGERED (IUCN) Facing an extremely high risk of extinction in the wild in the immediate future.
ENDANGERED (IUCN) Facing a very high risk of extinction in the wild in the near future.
VULNERABLE (IUCN) Facing a high risk of extinction in the wild in the medium-term future.
NEAR THREATENED (IUCN) Strong possibility of becoming endangered in the near future.
LEAST CONCERN (IUCN) Low-risk category that includes widespread and common species.
DATA DEFICIENT (IUCN) Not a threat category. Population and distribution data is insufficient for assessment.
NOT EVALUATED Data not yet assessed against the 5 IUCN criteria.
For fuller details see The IUCN Red List website (www.iucnredlist.org)

MAMMALS

Mammals, the most familiar group of vertebrates, all nourish their young on milk produced by the female's mammary glands (the unique skin structures after which the class is named). Most also give birth to live young and, with only a few exceptions, have a covering of hair on their body. Mammals are most widespread and diverse on land, but they have also colonized the water. Their success is largely due to their ability to maintain a constant internal body temperature, regardless of changing external conditions. Most mammals are also highly adaptable and often modify their behavior to suit changing circumstances. Some mammals, especially primates (the group that includes humans), form complex societies.

Evolution

The evolution of mammals was a gradual process that began in the late Paleozoic. The first mammals were small, shrew-like creatures that lived in the forests of the Carboniferous and Permian periods. They were characterized by a number of features that are still found in modern mammals, including a four-chambered stomach, a single lower jawbone, and a three-toed foot. Over time, mammals evolved a wide range of adaptations that allowed them to thrive in a variety of environments, from the deep sea to the highest mountains.

Anatomy

The anatomy of mammals is characterized by a number of key features. These include a four-chambered stomach, a single lower jawbone, and a three-toed foot. Mammals also have a unique skin structure called a mammary gland, which produces milk for their young. The skin of mammals is covered in hair, which provides insulation and protection. The internal organs of mammals are also highly specialized, with a complex system of blood vessels and a highly developed brain.

Reproduction

Mammals reproduce in a variety of ways. Some lay eggs, some give birth to live young, and some have a mix of both. The way an animal reproduces depends on its life cycle and its environment. Most mammals have a long gestation period and a long period of parental care for their young.

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